

The Carried Reference Tunes a Switch–Multiplier Continuum in Gated (GLU) MLPs

A companion note to *Transformer MLP Gate Thresholds Are Couplings to a Carried Reference Direction*

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Abstract

A prior paper showed that a trained transformer implements its MLP gate thresholds as a coupling $w \cdot b_L$ to a carried reference direction b_L , the corpus-mean layer-normalised gate input. We use that reference to give a functional reading of the gate in a gated (GLU) MLP. Each gate unit’s coupling to b_L places it on a continuous *switch–multiplier* spectrum: a unit strongly (negatively) coupled to b_L rests closed and behaves as a thresholded switch, while a unit orthogonal to b_L rests at the activation’s knee and, to first order, computes a bilinear product of its two branch reads. The per-unit cosine to b_L is the coordinate along that spectrum, and trained SwiGLU allocates units continuously across it, shifting toward the multiplier end with depth. Supporting this, the coupling localises to the gate branch (across four SwiGLU models the `gate_proj` rows align with b_L at 4–33× the `up_proj` rows) and lives in the reference’s high-variance subspace; a bit-identical-init training ladder shows the elementwise nonlinearity is what creates the switch end, and that among four feed-forward variants only the gated one spans both ends with independent key and value reads. We suggest this coverage, rather than any single mode, as a reason the GLU family is an effective default, and note the reading extends to the experts of a mixture-of-experts. The account is geometric and makes no quality claim.

1 Introduction

A companion paper (Tuomi, 2026) reported that a trained transformer implements its MLP gate thresholds as a coupling to a carried reference direction. The reference b_L is the corpus mean of the layer-normalised MLP input, the vector the gate rows read; a gate’s resting pre-activation decomposes exactly as $\langle w \cdot x_{\text{ln}} \rangle + b_n = w \cdot b_L + b_n$, and in the models studied the coupling $w \cdot b_L$ carries the resting inhibition of more than 99.9% of gates, at 50–60× the explicit bias parameter. Per read row the coupling is a thin slice of the row’s norm: factoring each row as $w = c_n \hat{b}_L + w_\perp$ (the anchor paper’s own notation), the fraction $|c_n|/\|w\|$ has median 0.07–0.12, so about 99% of every row is left free for content. We take this decomposition as established and build on it.

That paper reads the coupling as a threshold. A gated MLP invites a functional question the single-matrix picture does not: its read stage factorises into a gate branch (carrying the elementwise nonlinearity) and a linear value branch, whose product is written out, so a unit’s coupling to b_L can vary within a layer. We find it varies along one axis. Expanding the SiLU at its knee, a gate unit whose operating point $w_g \cdot b_L$ is deep-negative rests closed and fires only above threshold, a *switch*; a unit whose operating point sits at the knee ($w_g \perp b_L$) computes, to leading order, $\frac{1}{2}(w_g \cdot x)(w_v \cdot x)$, a bilinear *multiplier* of its two reads. A unit’s cosine to b_L is its coordinate on this switch–multiplier spectrum; trained SwiGLU spreads

units continuously along it, and the mix moves toward the multiplier end with depth. The rest of the note supports and bounds this reading: the coupling localises to the gate branch and to the reference’s high-variance subspace (§3); a controlled ladder shows the nonlinearity is what creates the switch end, and that of four feed-forward variants only the gated one spans the spectrum with independent key and value reads (§4); freeing the reference from the value branch returns a content dimension (§5); and we discuss why this coverage suits the GLU family, and how the reading should extend inside a mixture-of-experts (§6).

The claim is geometric, about where a shared rank-one direction sits in the read weights, and not a claim about quality: whether gating improves the loss is a separate question, addressed elsewhere at this feed-forward geometry (Shazeer, 2020), and we make no quality claim here. And what localises to the gate is the shared reference slice only: both branches carry content, so that freezing either one at its resting value does comparable damage to prediction (§7), as the product of two factors requires.

Throughout, “gate” is the branch carrying the elementwise nonlinearity ($w_g, \text{gate_proj}$), “value” the linear branch ($w_v, \text{up_proj}$); a SwiGLU MLP writes $W_{\text{down}}(\text{SiLU}(w_g x_{\text{in}}) \odot (w_v x_{\text{in}}))$. We use the reference $b_L := \langle x_{\text{in}} \rangle$ of the anchor paper: the corpus-mean layer-normalised MLP input, the direction the gates read. In the RMSNorm models x_{in} is the RMS-normalised input (no mean subtraction); b_L is well defined and is still what the gate rows dot.

2 The reference coupling places a unit on a switch–multiplier spectrum

A SwiGLU unit writes $\text{SiLU}(g) v d$ with $g = w_g \cdot x$, $v = w_v \cdot x$, and d its down-projection column. Expanding the SiLU about its knee, $\text{SiLU}(z) \approx \frac{1}{2}z + \frac{1}{4}z^2$, gives two regimes selected by where the unit rests, $g_0 = w_g \cdot b_L$:

- **Switch.** $g_0 \ll 0$: the gate rests closed, $\text{SiLU}(g) \approx 0$, and the unit fires only when the input pushes g above threshold, writing its value when it does. A thresholded, off-by-default detector.
- **Multiplier.** $g_0 \approx 0$ ($w_g \perp b_L$): the gate rests at the knee, $\text{SiLU}(g) v \approx \frac{1}{2}(w_g \cdot x)(w_v \cdot x)$, an always-on bilinear product of the two branch reads. This is the object a bilinear MLP is built from (Pearce et al., 2024).

The operating point g_0 , equivalently a unit’s cosine to b_L , is therefore a coordinate that slides the unit between the two regimes. (This is a per-unit coordinate set by the carried reference, not the global activation shape: SiLU/Swish interpolate linear-to-ReLU through a shared β (Ramachandran et al., 2017), whereas here each unit’s regime is set by its own alignment to b_L .)

Trained SwiGLU populates the whole spectrum. Table 1 splits gate units at three layers of Qwen2.5-3B into the switch end (coupling above $2\times$ the random-alignment floor) and the multiplier end (below it), and reports each group’s *duty* – the anchor paper’s term for the fraction of positions at which a unit’s pre-activation is positive. The switch units rest deep and fire rarely (duty near 0), with a pre-activation dominated by the constant operating point (ratio of $|\text{mean}|$ to standard deviation well above one); the multiplier units rest near the knee, fire about a third of the time (duty $\approx 1/3$), and are input-driven (that ratio near zero). The coupling is graded rather than bimodal (the cosine distribution is a smooth tail, no gap), predominantly inhibitory in sign (88–100% of gate units), and scales with the row norm (Spearman 0.55–0.66); the multiplier fraction rises with depth (4% $\rightarrow$ 31% $\rightarrow$ 39% here). The multiplier-end units are input-driven content gates, not a second reference: their rows share no dominant common axis (top singular value $\leq 1.6\%$ of energy) and are not aligned with the value branch ($\cos(w_g, w_v) \approx 0$).

These labels are behavioural, not only geometric. Comparing each unit’s actual output $\text{SiLU}(g) v$ to the leading bilinear term $g v$ on tokens, the knee (uncoupled) units are well approximated by the product –

L	mode	share	rest $w_g \cdot b_L$	duty	mean /sd
9	switch	96%	−4.79	0.00	3.2
	multiplier	4%	−0.73	0.19	0.8
18	switch	69%	−0.62	0.13	1.1
	multiplier	31%	−0.22	0.32	0.4
27	switch	61%	−0.87	0.11	1.3
	multiplier	39%	−0.23	0.34	0.4

Table 1: Gate units at the two ends of the spectrum (Qwen2.5-3B), split by coupling to b_L . Switch units rest deep-negative, fire rarely, and are constant-dominated; multiplier units rest near the knee, fire $\sim 1/3$ of the time, and are input-driven. The multiplier share grows with depth. Source: `census_gated_coupling_dist.py`, `census_gated_uncoupled_units.py`.

L	$ \cos(w_g, \hat{b}_L) $	$ \cos(w_v, \hat{b}_L) $	ratio	resting SiLU($w_g \cdot b_L$)	$\cos(w_g, w_v)$
9	0.0799	0.00232	34.4	−0.042	0.000
18	0.047	0.0056	8.4	−0.180	0.001
27	0.044	0.0062	7.1	−0.204	−0.001

Table 2: Gate vs. value read geometry, Qwen2.5-3B (weights and b_L only, mid-stack layers of 36). Gate rows align with the reference 7–34 $\times$ the value rows; the gate rests closed; the two branches are mutually orthogonal (not self-gating). Gate high-variance-coordinate energy exceeds the value’s throughout (0.03–0.04 vs. 0.015 against an isotropic 0.031). Source: `census_gated_gate_structure.py`, Leg B.

median R^2 0.55–0.83 and best-fit slope ≈ 0.4 , near the knee’s coefficient of $\frac{1}{2}$ – so their output genuinely is the bilinear term. The deeply-coupled switch units are not: at the shallowest layer, where switches rest deepest, their R^2 falls to 0.15 with slope ≈ 0 , the behaviour of a threshold rather than a product. The bilinear fidelity decreases with coupling at every layer (Spearman between R^2 and $|\cos(w_g, \hat{b}_L)|$ is -0.17 to -0.42), so a unit’s geometric coordinate predicts whether it actually computes a product. The gap is sharp at the extremes and graded between them, where softly-resting units track the product either way – the signature of a continuum rather than two classes.

3 The coupling localises to the gate branch

The switch end of the spectrum lives in the gate branch, and its geometry is visible in the weights alone. For each hidden unit we take the cosine of its gate row w_g and its value row w_v with the unit reference $\hat{b}_L$, normalised by the row norm, and read the median over units at three mid-stack layers of Qwen2.5-3B (Table 2). The gate rows carry the coupling and the value rows do not: $|\cos(w_g, \hat{b}_L)|$ is 34 $\times$ the value figure at L9 and 7–8 $\times$ deeper, a dissociation of the same quantity the companion paper measured inside a single matrix, now split across the two projections. The sign is consistent with a threshold: the gate’s resting pre-activation $w_g \cdot b_L$ is negative, so the resting gate is closed ($\text{SiLU}(w_g \cdot b_L) < 0$ at every layer) and opens conditionally. The two branches are mutually orthogonal ($\cos(w_g, w_v) \approx 0$), each reading a distinct direction, and the gate rows place more of their energy on the high-variance coordinates of x_{in} than the value rows do (0.03–0.04 against 0.015, against an isotropic 0.031), the first sign that the coupling lives in the amplitude part of the reference, which the control below makes precise. A forward pass agrees, weakly: the gate activation tracks $\|x_{\text{in}}\|$ across positions at 1.2–1.9 $\times$ the value’s correlation, strongest at L9.

model	gate/value $ \cos(\cdot, \hat{b}_L) $ ratio			resting SiLU
	$L \approx .25$	$L \approx .5$	$L \approx .75$	
Qwen2.5-1.5B	11.6	7.3	8.0	all < 0
Qwen2.5-3B	33.0	8.1	7.1	all < 0
TinyLlama-1.1B	4.4	5.2	4.7	all < 0
SmolLM2-1.7B	5.3	4.8	1.8	all < 0

Table 3: Cross-model replication (weights $+b_L$, three mid-stack layers each). The reference coupling sits in the gate branch in every model; the gate rests closed everywhere. It weakens at SmolLM2’s deepest sampled layer (ratio 1.8, gate frac-open 0.34), the clearest soft spot; splitting SmolLM2’s own gate units by the switch/multiplier threshold of Table 1 confirms a population shift toward the multiplier end there (30% $\rightarrow$ 41% $\rightarrow$ 60% from shallowest to deepest sampled layer). Sources: `census_gated_branch_xmodel.py`, `census_gated_uncoupled_smolllm2.py`.

Where in the reference the coupling sits. On Phi-2 the companion paper finds mid-stack b_L to be a distributed direction, distinct from the massive-activation and outlier-coordinate objects (Tuomi, 2026). The RMSNorm SwiGLU models here are the other case: b_L concentrates, with the eight highest-variance coordinates carrying 66–75% of $\|b_L\|^2$ across the three layers. This concentration reflects the model’s massive-activation profile rather than its normalisation scheme: across seven models of both normalisation types, b_L ’s largest coordinates coincide with the massive-activation coordinates and its concentration tracks their strength, which is why the same reference reads as distributed in Phi-2 and concentrated here (Sun et al., 2024). Either way, whether the gate reads the reference or merely those coordinates has to be settled again for these models, and for the branch split in particular. Zeroing the eight coordinates and renormalising, so that two-thirds of the reference is discarded, the gate still aligns with what remains at $12.8/4.0/3.2 \times$ the value (L9/L18/L27) and still rests closed. The localisation narrows as more of the high-variance subspace is removed: with the top sixty-four coordinates gone the ratio falls to about 1.3 and the value’s alignment rises to meet the gate’s in the low-variance bulk. The gate carries the reference coupling within the reference’s high-variance subspace, the same subspace the energy reading above points to.

Replication across SwiGLU lineages. The same static reading on three mid-stack layers of four SwiGLU models (Table 3) puts the coupling in the gate in every one. The gate rows align with b_L more than the value rows at all twelve model-layers, and the gate rests closed (negative resting SiLU) everywhere; the strength varies by lineage, with Qwen sharpest. The clearest soft spot is SmolLM2’s deepest sampled layer, where the ratio falls to 1.8 and the gate opens at a third of positions rather than a tenth. Splitting SmolLM2’s own gate units by the same switch/multiplier threshold as Table 1 confirms this is a population shift, not an unexplained dip: the multiplier-end share rises 30% $\rightarrow$ 41% $\rightarrow$ 60% from the shallowest to the deepest sampled layer, the same monotonic climb as Qwen2.5-3B’s 4% $\rightarrow$ 31% $\rightarrow$ 39% (§2), only starting and ending further along the spectrum. Across the other eleven layers the separation is unambiguous, so the localisation is a property of the SwiGLU construction and not of one model, with each model’s multiplier-end mixture setting how far it softens.

pair (edit)	step 0	step 12k	step 150k
swiglu vs. bilinear (add a gate)	0.89	0.22	0.17
gelu vs. relu2 (swap elementwise fn)	0.97	0.89	0.74

Table 4: Cross-arm cosine of the reference b_L between matched arms trained from bit-identical initialisation (median over 12 layers). Adding a gate relocates b_L ; swapping the elementwise function does not. Both pairs start aligned; the divergence is trained. At step 20000 a double-gated arm sits at 0.72 from swiglu and 0.18 from bilinear. Source: `ladder_gate_reference.py`.

4 The nonlinearity creates the switch end, and only the gated arm spans the spectrum

The switch end of the spectrum is the part that needs a carried reference: a bare product has no off-by-default point. To see that the elementwise nonlinearity is what supplies it, and which architectures can reach both ends, we use a controlled training ladder. Four feed-forward variants were trained at the 160M (Pythia) scale on identical data in identical order under a shared schedule, differing only in the feed-forward block: a plain GELU MLP, a SwiGLU, a bilinear GLU (the product of two linear branches, with no elementwise nonlinearity), and a squared-ReLU MLP. The matched-shape arms share bit-identical initialisation (maximum absolute weight difference 0): the SwiGLU gate and value rows are the bilinear arm’s two branches at initialisation, and the GELU read matrix is the squared-ReLU arm’s. The SwiGLU-versus-bilinear pair therefore isolates a single edit, adding the SiLU to one branch, and the GELU-versus-squared-ReLU pair isolates swapping the elementwise function, each with everything else fixed.

Taking the cross-arm cosine of b_L at matched layers (Table 4), adding the gate relocates the reference while swapping the elementwise function leaves it in place: the SwiGLU–bilinear cosine falls to 0.17 over training while the GELU–squared-ReLU cosine holds at 0.74.¹ Both pairs begin aligned at initialisation (0.89 and 0.97), so the divergence is trained rather than a starting condition. The mechanism is the one the mature models show: over the same training the SwiGLU gate develops its gate-over-value alignment ratio (rising to about 13 mid-stack) and its resting gate closes, while the bilinear arm’s two symmetric branches stay at ratio 1 with no closed rest, a built-in null that differs only by the SiLU. The SiLU places the coupling in the gate, and placing it there is what moves the reference.

The double-gated arm. A third gated arm, with the SiLU on both branches (again bit-identical initialisation to the SwiGLU), shows where the coupling goes when two branches carry a threshold rather than one: it splits between them in equal shares. The two branches sit at alignment ratio 1, both rest closed, and each carries b_L as its top read principal component at about half the single-gate strength; the double-gated arm’s reference sits at cosine 0.72 from the SwiGLU and 0.18 from the bilinear. Across the three gated arms the pattern is that the reference coupling distributes over every branch that carries a threshold, in equal shares by symmetry: one gate takes the coupling and leaves the value branch free, two gates split it and free neither, and the ungated bilinear product anchors it to neither branch. This arm is measured at a single training checkpoint.

¹ b_L is the mean post-LayerNorm MLP input, matching the anchor’s definition. Measuring the mean pre-normalisation residual instead gives the same dissociation at a wider spread (0.12 versus 0.93), so it is not an artifact of the reference’s definition.

arm	rest (med)	off-by-default	at knee	key/value decoupled
gelu	−2.20	92%	1%	no
relu2	−1.45	88%	3%	no
swiglu	−0.84	28%	18%	yes
bilinear	+0.00	0%	96%	yes

Table 5: Operating-point spectrum across the ladder arms (step 150000, gate/read branch, pooled over three mid-stack layers). Off-by-default is the fraction of units with duty < 0.15 ; at-knee is duty in $[0.35, 0.65]$. Only the gated SwiGLU carries mass at both ends with independent branches. Source: `census_ladder_gate_spectrum.py`.

Which architectures reach which ends. Reading the operating point of each arm’s gate/read branch at step 150000 (Table 5) places the four architectures on the same axis. The non-gated arms (GELU, squared-ReLU) are almost entirely switches: 88–92% of their units rest off-by-default, but a single read matrix ties key to value, so they cannot occupy the multiplier end with independent reads. The bilinear arm is the mirror image: with no off-state, 96% of its units sit at the knee and none rest closed, so it reaches only the multiplier end. Only SwiGLU carries mass at both ends (28% off-by-default, 18% at the knee, the rest in between) with a wide coupling spread and two independent branches. Reading the median resting point down the arms gives a monotone march from all-switch (GELU -2.2) to all-multiplier (bilinear 0.0), with SwiGLU between them. So the gated architecture is the one that spans the switch–multiplier spectrum with a decoupled key and value.

5 Freeing the reference returns a content dimension

The reference coupling is a rank-one overhead shared by every read row: the same direction $\hat{b}_L$ enters each row’s decomposition, so across the read matrix it occupies one shared axis rather than per-row capacity. In the non-gated arms that axis is the read matrix’s top principal component (Table 6): b_L aligns with the first singular direction of the GELU and squared-ReLU read matrices at $|\cos| \approx 0.85$, carrying read energy about $17\text{--}20\times$ the isotropic share. SwiGLU moves that axis into the gate, where b_L is again the top component, and leaves the value branch without it: there b_L is only the matrix’s 78th principal component, at the isotropic floor. The value branch is a content read matrix with the shared reference removed, and removing it returns the branch’s top principal direction to content. The recovered capacity is real but small on this measure – the value branch’s effective rank is about 1.5% of d above the GELU read matrix’s, matching a GELU matrix with b_L projected out. This sharpens the anchor’s per-row coupling figure, locating the shared slice as the top read component rather than a diffuse per-row tax; it is a structural statement, not a performance gain, which this setup does not measure.

6 Discussion

A reason the GLU family is an effective default. The spectrum suggests why gating is a safe architectural choice, as a coverage argument rather than a quality one. A language model uses both thresholded detectors and multiplicative features; one GLU unit type reaches either, its position set after the fact by training, whereas the bilinear MLP reaches only the multiplier end (no off-state) and a single ReLU/GELU unit reaches only the switch end (a multiplier needs about two units, by the polarisation identity $ab = \frac{1}{4}[(a+b)^2 - (a-b)^2]$). The gated unit and the bilinear MLP spend the same per-unit budget

arm : branch	tax	shared/iso	b_L PC#	PC cos
gelu : read	0.15	19.7	1	0.85
relu2 : read	0.14	16.4	1	0.83
swiglu : gate	0.11	14.0	1	0.71
swiglu : value	0.02	1.7	78	0.32
bilinear : a/b	0.03	~ 2.5	2	0.40

Table 6: Where the shared reference sits in the read matrix (ladder, step 150000, medians over layers). In non-gated arms b_L is the top principal component of the single read matrix; SwiGLU moves it into the gate and leaves the value branch’s top component free for content. Source: `ladder_content_capacity.py`.

– two reads and a down-projection write, $3d$ parameters – but only the gated unit covers both ends, because its SiLU sits on exactly one of the two reads: that read’s coupling to b_L can rest the unit closed, which neither of the bilinear MLP’s two linear reads can do, so it always multiplies. A single ReLU/GELU unit spends less ($2d$: one read, one write) and is pinned to the switch end for the complementary reason – it has no second read to multiply against. This is a capability statement, not a prediction of lower loss; why our ladder cannot adjudicate one, and where the loss comparison at this feed-forward geometry is made instead, is in §7. The coverage argument speaks to what each architecture can represent, not to a measured win.

Every trained unit uses both reads. One thing the measurements settle is that the switch–multiplier split is not a split into single-read and two-read units. Trained SwiGLU has independent key and value reads on essentially every unit ($\cos(w_g, w_v) \approx 0$ throughout), switch-end units included, so a unit’s coupling to the reference sets its operating point, not whether it uses its second read. That value read earns its keep across the whole population, not only at the multiplier end, which is the substance behind the $3d$ -versus- $2d$ accounting GLU implementations use to match a non-gated MLP’s width. Oskin’s NC-FFN (§8), a GELU highway with a minority of multiplicative operand units (Oskin, 2026), ties GELU, a fully-multiplicative version fails to train, and its operators prefer a soft, graded regime whose crisp two-operand logic erodes over training (his §7.6). That preference matches the drift toward the soft, near-knee end we observe, with one difference worth stating: his imposed second operand decays toward single-operand gating, whereas SwiGLU’s value branch stays informative (§7), so the soft units here remain genuine two-read multipliers.

The reading extends inside mixture-of-experts experts. Each expert in an MoE is itself a gated MLP, and an expert sees only the tokens routed to it, so its carried reference should be the mean over its *own* routed tokens rather than a global mean. A pilot on OLMoE-1B-7B (64 experts, top-8; three mid-stack layers) bears this out with two qualifications. The per-expert reading holds: within each expert the coupling localises to the gate branch (gate/value alignment ratio 4.0/2.3/1.7, weakening with depth as in the dense soft spot), and each expert’s gate rests closed on its *own* routed reference ($\text{SiLU}(w_g \cdot b_L^{(e)}) < 0$). The reference is expert-conditional but shared-dominated: an expert’s routed mean sits at cosine 0.79–0.86 to the global mean and 0.61–0.73 to other experts’ references, a per-expert tilt on a common reference that grows with depth. Most cleanly, the routed reference is essentially orthogonal to the expert’s router direction ($\cos \approx 0.00$ –0.03): the computation-stage reference and the selection-stage routing vector are distinct objects, which makes the separation from Li et al. (2026) – who project the gate weight onto the router direction as a per-token confidence bias – an empirical finding rather than only a conceptual one.

A second pilot on Qwen1.5-MoE-A2.7B (60 routed experts top-4 plus one shared expert) adds the contrast OLMoE lacks. The shared expert, which processes every token, reads the most like a dense MLP against the global reference: its gate/value coupling ratio ($2.5\text{--}10.6\times$ across the three layers) exceeds the routed experts' ($1.4\text{--}3.7\times$), it aligns to the global reference more than they do, and it rests closed. The routed experts instead couple to their own tilted references (cosine $0.86\text{--}0.88$ to the global mean, $0.71\text{--}0.78$ across experts), and the routed-reference-versus-router-direction cosine is again small (here mildly negative, -0.20 to -0.27 , against ≈ 0 in OLMoE), far from the value coincidence would give. So across two MoE families the always-on expert carries the global reference while routed experts specialise to their routed ones, and the two-level separation holds. These are single-seed pilots on two models; the depth and cross-family details are not pinned down.

7 Scope and limitations

- **Both branches carry content.** What localises to the gate is the shared reference slice, and only that. The gate branch carries content as well: its row is the coupling $c_n \hat{b}_L$ plus a content remainder $w_\perp$ that is about 99% of the row, exactly as in the anchor's per-row anatomy. Freezing either branch at its resting value does comparable damage to prediction (value-over-gate top-1 flip $0.84\text{--}1.18$ and ΔNLL $0.5\text{--}1.8\times$ across the three layers), as a product of two factors requires.
- **The coupling sits in the reference's high-variance subspace.** As §3 reports, the localisation survives removing the eight massive coordinates (ratio $3\text{--}13\times$) but narrows toward parity once the top sixty-four variance coordinates are gone. The claim is a high-variance-subspace coupling, not a coupling to the reference as a whole.
- **The multiplier reading is leading-order.** A knee-resting unit is bilinear only to first order in the SiLU expansion; the curvature term and finite pre-activation add higher-order structure, so it approximates rather than equals a product. The within-model spectrum (Table 1) is a single model (Qwen2.5-3B); the cross-architecture span (Table 5) is the 160M ladder.
- **Quality is out of scope, and our ladder cannot speak to it.** The ladder is 160M and trained only to a few billion tokens, so it grounds the geometric relocation but is far too small and short to adjudicate loss differences; we draw no quality conclusion from it, in either direction. The loss comparison at this feed-forward geometry is elsewhere (Shazeer, 2020); our claim is about where a shared reference sits in the two branches.
- **Two seeds on the ladder relocation.** The relocation reproduces at a second initialisation: at step 12000 (about 80% of the effect) the SwiGLU–bilinear cosine is 0.30 against the GELU–squared-ReLU control's 0.84 , the same dissociation as the first seed (0.22 versus 0.89 at that step). The full-horizon replicate and further seeds are not run.
- **The reference is input-conditioned and rotates across depth.** Inherited from the anchor: b_L shifts with the input distribution (prose to code cosine $0.76\text{--}0.84$, orthogonal for a degenerate repeated token) and rotates slowly across the stack. Every statement here is per layer and for the reading distribution rather than about a single global direction.

8 Related work

This note extends Tuomi (2026): that paper establishes the coupling $w \cdot b_L$ within a single read matrix, and the result here is that in a gated MLP the coupling sits in the gate branch while the value branch is free of it, a two-branch contrast the single-matrix treatment does not make.

Gated linear units were introduced by Dauphin et al. (2017); Shazeer (2020) evaluates GLU variants, including SwiGLU and the bilinear MLP, in the Transformer feed-forward layer, finding them at least matching GELU and offering no account of why; the object here is functional, where each unit sits on the switch–multiplier spectrum. The multiplier end is the bilinear MLP that Pearce et al. (2024) analyse as a product with no elementwise nonlinearity; our reading places ordinary SwiGLU units on a continuum whose far end is that bilinear object, indexed by coupling to b_L . The nearest architecture that assigns the two roles to separate unit types by construction is Oskin’s NC-FFN (Oskin, 2026); we compare its trained behaviour to the depth-wise drift we measure once the switch–multiplier coordinate is defined, in §6. The global activation shape gives a different, shared interpolation: SiLU/Swish move from linear to ReLU through one parameter β (Ramachandran et al., 2017), whereas our coordinate is per unit and set by the carried reference. The closest mechanism is Li et al. (2026), who split an expert gate weight into a component along the router direction (a per-token “confidence” bias that slides a unit along the SiLU curve) and an orthogonal remainder; we differ in the reference (a carried corpus-mean, not a learned router vector), the setting (trained dense models, not a proposed MoE architecture), the coordinate (per unit and static, not per token), and in naming the multiplier pole. Weight-based taxonomies of gated neurons are adjacent: Gerstner and Schütze (2025) classify units by input–output weight cosine with a depth trend (a different cosine, measuring residual read/write alignment) and Gerstner and Schütze (2026) by the four gate/value sign regimes; neither uses a carried reference. Finally, massive activations have been read as carried near-constant components with a bias-like role (Sun et al., 2024) and MLP units as binary routers (Balogh, 2026); the reading here is consistent with both, and §3 identifies the reference-carrying coordinates as the massive ones.

Reproducibility

Scripts and JSON artifacts (fixed seeds, public checkpoints where applicable):

`census_gated_gate_structure.py` (Table 2, gate/value geometry),
`census_gated_massive_control.py` (the massive-coordinate control, §3),
`census_gated_branch_xmodel.py` and `census_gated_uncoupled_smollm2.py`
(Table 3), `census_bL_shape_xarch.py` (the cross-architecture b_L -shape measurement, §3),
`census_gated_coupling_dist.py` and `census_gated_uncoupled_units.py` (Table 1,
the switch–multiplier split and the content-gate characterisation),
`census_gate_bilinear_fidelity.py` (the behavioural bilinear-fidelity check, §2),
`census_gated_value_content.py` (§7), `ladder_gate_reference.py` (Table 4),
`census_ladder_gate_spectrum.py` (Table 5) and `ladder_content_capacity.py`
(Table 6), with the feed-forward variants in `ffn_variants.py`. The training ladder is the
bit-identical-initialisation four-arm set of Tuomi (2026)’s companion GLU study; the seed replicate uses
`run_glu_matched.py` with a second initialisation and `measure_seed1_relocation.py`.
The mixture-of-experts pilots are `census_moe_expert_reference.py` (OLMoE) and
`census_moe_shared_vs_routed.py` (Qwen1.5-MoE), §6. Code and data are available at
https://github.com/EvidentSolutions/llm-interp/tree/main/background_glu and

archived on Zenodo.

Use of AI Assistants

Large language models were used as assistive tools for coding, running experiments, and drafting text. All research questions, experimental design, and reported claims were directed and verified by the author.

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